Python Programming Lab. [Batch-B3]

Assignment Submission Details

Field	Details
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Course Name	Python Programming Lab. (MCA31PC06)
Academic Year	2025-26 (Semester-1)
Course Teacher	Prof. Prakash Ukhalkar
Assignment Name	Assignments based on Input Output and File Handling using Python 1.
Assignment Number	Assignment 10
Submission Date *	27-11-2025

1.Write a Python program to read an entire text file..

In [2]:
```
try:
    with open("sample.txt", "r") as file:
        content = file.read()
        print(content)

except FileNotFoundError:
    print("Error: sample.txt not found")
```

```
Error: sample.txt not found
```

In [12]:
```
%%writefile sample.txt
Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that coun
The future belongs to those who prepare for it today.
Don't watch the clock; do what it does. Keep going.
Dream big. Work hard. Stay focused.
```

```
Writing sample.txt
```

In [14]:
```
try:
    with open("sample.txt", "r") as file:
        content = file.read()
        print("Data in sample.txt")
        print(content)

except FileNotFoundError:
    print("Error: sample.txt not found")
```

```
Data in sample.txt
Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that count
s.
The future belongs to those who prepare for it today.
Don't watch the clock; do what it does. Keep going.
Dream big. Work hard. Stay focused.
```

2.Write a Python program to append text to a file and display the text.

In [18]:
```python
try:
    with open("sample.txt", "a") as file:
        file.write("\nThis is the appended text to sample.txt\n")
        file.write("Believe in yourself and all that you are.\n")
        file.write("Every day is a second chance.\n")
        file.write("Push yourself, because no one else is going to do it for you.\n

    with open("sample.txt", "r") as file:
        print(file.read())

except FileNotFoundError:
    print("Error: sample.txt not found.")
```

```
Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that count
s.
The future belongs to those who prepare for it today.
Don't watch the clock; do what it does. Keep going.
Dream big. Work hard. Stay focused.


This is the appended text to sample.txt
Believe in yourself and all that you are.
Every day is a second chance.
Push yourself, because no one else is going to do it for you.
```

3.Write a Python program to read a file line by line and store it into an array..

In [31]:
```python
data = []

try:
    with open("sample.txt", "r") as file:
        for line in file:
            data.append(line)   #without using strip()
    print("data in sample.txt without using strip()\n")
    for item in data:
        print(item)

except FileNotFoundError:
    print("Error: sample.txt not found.")
```

```
data in sample.txt without using strip()

Believe you can and you're halfway there.

Success is not final, failure is not fatal: it is the courage to continue that count
s.

The future belongs to those who prepare for it today.

Don't watch the clock; do what it does. Keep going.

Dream big. Work hard. Stay focused.

            This is the appended text to sample.txt

Believe in yourself and all that you are.

Every day is a second chance.

Push yourself, because no one else is going to do it for you.
```

In [32]:

```python
data = []

try:
    with open("sample.txt", "r") as file:
        for line in file:
            data.append(line.strip())   # using strip()
#strip() is used to remove extra spaces and the newline \n at the start or end of a
    print("data in sample.txt using strip()\n")
    for item in data:
        print(item)

except FileNotFoundError:
    print("Error: sample.txt not found.")
```

```
data in sample.txt using strip()

Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that count
s.
The future belongs to those who prepare for it today.
Don't watch the clock; do what it does. Keep going.
Dream big. Work hard. Stay focused.
This is the appended text to sample.txt
Believe in yourself and all that you are.
Every day is a second chance.
Push yourself, because no one else is going to do it for you.
```

4.Write a python program to find the longest words.

In [ ]:

```python
try:
    with open("sample.txt", "r") as file:
        content = file.read()
        print("Content in sample.txt:\n",content)
```

```
        words = content.split()


    print("After using split(), words list contains:\n", words)

    max_len = max(len(word) for word in words)

    #here first we find the length of each word in list and apply max() on it and s

    longest_words = [word for word in words if len(word) == max_len]
    #check length of each word if it equal to max_len then we store it in word
    print("\nLongest words:", longest_words)

except FileNotFoundError:
    print("Error: The file 'sample.txt' was not found.")
```

```
Content in sample.txt:
 Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that count
s continue
here we write some words having same length continue continue continue continue

After using split(), words list contains:
 ['Believe', 'you', 'can', 'and', "you're", 'halfway', 'there.', 'Success', 'is', 'n
ot', 'final,', 'failure', 'is', 'not', 'fatal:', 'it', 'is', 'the', 'courage', 'to',
'continue', 'that', 'counts', 'continue', 'here', 'we', 'write', 'some', 'words', 'h
aving', 'same', 'length', 'continue', 'continue', 'continue', 'continue']

Longest words: ['continue', 'continue', 'continue', 'continue', 'continue', 'continu
e']
```

5.Write a Python program to create a file where all letters of the English alphabet are listed by a specified number of letters on each line.

In [28]:

```
import string

n = int(input("Enter number of letters per line: "))
print(f"user wants {n} letters on each line of file")
letters = string.ascii_lowercase    # abcdefghijklmnopqrstuvwxyz

with open("sample2.txt", "w") as file:
    for i in range(0, len(letters), n):

        # here i takes values: 0, n, 2n, 3n, ...
        #assume n=4
        #here i can have values 0 4 8 12 16 20 24

        file.write(letters[i:i+n] + "\n")

        #means on first line of file letters[0:4] i.e abcd
        #on second line of file letters[4:8] i.e efgh

print("sample2.txt file created successfully")
with open("sample2.txt", "r") as file:
```

```
    print("Content of sample2.txt:\n")
    print(file.read())
```

```
user wants 4 letters on each line of file
sample2.txt file created successfully
Content of sample2.txt:

abcd
efgh
ijkl
mnop
qrst
uvwx
yz
```

6.Implement a program that opens a file in write mode, but only if it doesn't already exist

In [30]:
```
import os

print("\nFiles in current directory:")
for file in os.listdir():
    print(file)

filename = "newfile.txt"

if not os.path.exists(filename):
    with open(filename, "x") as file:
        file.write("\nFile created successfully")
    print("File created:", filename)
else:
    print("File already exists")
```

```
Files in current directory:
.dist
125M1H055_A02_PPLAB_B3.html
125M1H055_A02_PPLAB_B3.ipynb
125M1H055_A04_PPLAB_B3.html
125M1H055_A04_PPLAB_B3.ipynb
125M1H055_A04_PPLAB_B3.pdf
125M1H055_A05_PP_LAB_B3.html
125M1H055_A05_PP_LAB_B3.ipynb
125M1H055_A05_PP_LAB_B3.pdf
125M1H055_A06_PP_LAB_B3.html
125M1H055_A06_PP_LAB_B3.ipynb
125M1H055_A06_PP_LAB_B3.pdf
125M1H055_A07_PP_LAB_B3.html
125M1H055_A07_PP_LAB_B3.ipynb
125M1H055_A07_PP_LAB_B3.pdf
125M1H055_A08_PP_LAB_B3.html
125M1H055_A08_PP_LAB_B3.ipynb
125M1H055_A08_PP_LAB_B3.pdf
125M1H055_A09_PP_LAB_B3.html
125M1H055_A09_PP_LAB_B3.ipynb
125M1H055_A09_PP_LAB_B3.pdf
125M1H055_A10_PP_LAB_B3.ipynb
125M1H055_A11_PP_LAB_B3.ipynb
125M1H055_A12_PP_LAB_B3.ipynb
125M1H055_A13_PP_LAB_B3.ipynb
125M1H055_A14_PP_LAB_B3.ipynb
125M1H055_A15_PP_LAB_B3.ipynb
adminonly.txt
assignments.txt
basics.py
custom_exceptions.py
Exception_Handling.pdf
File_Handling.pdf
inbuilt_modules_in_python.pdf
learn-python-from-scratch
math_umodule.py
modules_in_python.pdf
myfolder
myfolder2
q1.ipynb
q2.ipynb
q3.ipynb
sample.txt
sample2.txt
__pycache__
File created: newfile.txt
```

7.Write code to read a CSV file containing product information and create a list of product objects

In [ ]:
```
#to read csv file we need csv file so we create it
import csv

data = [
```

```
    ["ID", "Name", "Price"],
    ["201", "Tablet", "15000"],
    ["202", "Charger", "600"],
    ["203", "HDMI Cable", "350"]
]

with open("products.csv", "w", newline="") as file:
    #newline="" prevents writing extra new line into file
    writer = csv.writer(file)
    writer.writerows(data)

print("products.csv created successfully")
```

```
products.csv created successfully
```

In [34]:

```
import csv

product_list = []

try:
    with open("products.csv", "r") as file:
        reader = csv.reader(file)

        for row in reader:
            product_list.append(row)

    print("Data in products.csv:\n")
    for product in product_list:
        print(product)

except FileNotFoundError:
    print("Error: products.csv not found.")
```

```
Data in products.csv:

['ID', 'Name', 'Price']
['201', 'Tablet', '15000']
['202', 'Charger', '600']
['203', 'HDMI Cable', '350']
```

8.Create a program that generates a CSV report from a list of dictionaries..

In [ ]:

```
import csv

products = [
    {"ID": "201", "Name": "Tablet", "Price": "15000"},
    {"ID": "202", "Name": "Charger", "Price": "600"},
    {"ID": "203", "Name": "HDMI Cable", "Price": "350"}
]

with open("report.csv", "w", newline="") as file:
    writer = csv.DictWriter(file, fieldnames=products[0].keys())
    # DictWriter is CSV writer used for dictionaries
    # products[0].keys() gives ['ID', 'Name', 'Price']

    writer.writeheader()        # writes: ID, Name, Price
```

```
    writer.writerows(products)  # writes product rows

print("report.csv generated\n")


print("Content of report.csv\n")

with open("report.csv", "r") as file:
    print(file.read())
```

```
report.csv generated

Content of report.csv:

ID,Name,Price
201,Tablet,15000
202,Charger,600
203,HDMI Cable,350
```

9. Write a Python program that reads the contents of a text file and prints the number of lines, words, and characters in the file.

In [ ]:
```
try:
    with open("sample.txt", "r") as file:
        content = file.read()
    print(content)
    lines = content.count("\n") + 1
    # add 1 because no. of lines = no. of newline characters + 1
    #last line does not contain \n character
    words = len(content.split())
    chars = len(content)

    print("Number of lines:", lines)
    print("Number of words:", words)
    print("Number of characters:", chars)

except FileNotFoundError:
    print("Error: sample.txt not found.")
```

```
Believe you can and you're halfway there.
Success is not final, failure is not fatal: it is the courage to continue that count
s continue
here we write some words having same length continue continue continue continue.
Number of lines: 3
Number of words: 36
Number of characters: 218
```

10. Create a program that takes user input to write a series of lines to a new text file. Ensure that each line is written to the file on a new line.

In [42]:
```
filename = "notes.txt"

with open(filename, "w") as file:
    print("Enter lines to write to the file. Enter 0 to stop.\n")
```

```python
    while True:
        line = input("Enter a line: ")
        if line == "0":
            break
        file.write(line + "\n")   # write each line on a new line

print("Data written to", filename)


print("Data of notes.txt:\n")

with open(filename, "r") as file:
    print(file.read())
```

```
Enter lines to write to the file. Enter 0 to stop.

Data written to notes.txt
Data of notes.txt:

this is the first line
this is the second line
this is the third line
this is the last line of notes.txt
```